

Water Segmentation using Multispectral and Optical Data (First Week)

Objective:

Develop a robust solution for accurately segmenting water bodies using multispectral and optical data. This solution is vital for monitoring water resources, flood management, and environmental conservation, where precise segmentation can significantly impact decision-making.

Tasks

- Understand the dataset structure: number of samples, image shape, channel/band layout, and label format.
- Visualize all 12 spectral bands individually to develop intuition for what each band captures.
- Explore the binary mask labels and understand the positive/negative class distribution.
- Research multispectral segmentation: how 12-channel input differs from RGB, what bands are most informative for water detection. (Note the 12 band contains the RGB, but each sensor of the rest describe specific information.)
- Research the U-Net architecture: encoder-decoder structure, skip connections, and why it suits pixel-level segmentation.
- Preprocess the 12-channel data: apply per-band normalization and verify the output shape is (12, 128, 128).
- Implement U-Net from scratch in PyTorch supporting 12-channel input and binary mask output.
- Train the model and evaluate using IoU, precision, recall, and F1-score on the water class.

Completion Requirements

- Dataset exploration notebook with band visualizations
- Preprocessing pipeline correctly normalizes all 12 bands and outputs tensors of shape (12, 128, 128).
- U-Net model is implemented from scratch (no pretrained weights) with 12-channel input and binary output.
- Model trains without errors and produces a validation IoU reported at the end of training.
- Code is pushed to a GitHub repository with a clear README

Resources

- <https://arxiv.org/abs/1505.04597>
- <https://support.esri.com/en-us/gis-dictionary/near-infrared>
- <https://www.sciencedirect.com/science/article/abs/pii/S0924271619301522>
- <https://custom-scripts.sentinel-hub.com/custom-scripts/sentinel-2/swir-rgb/>
- <https://www.usgs.gov/landsat-missions/landsat-8-9-surface-reflectance-quality-assessment>